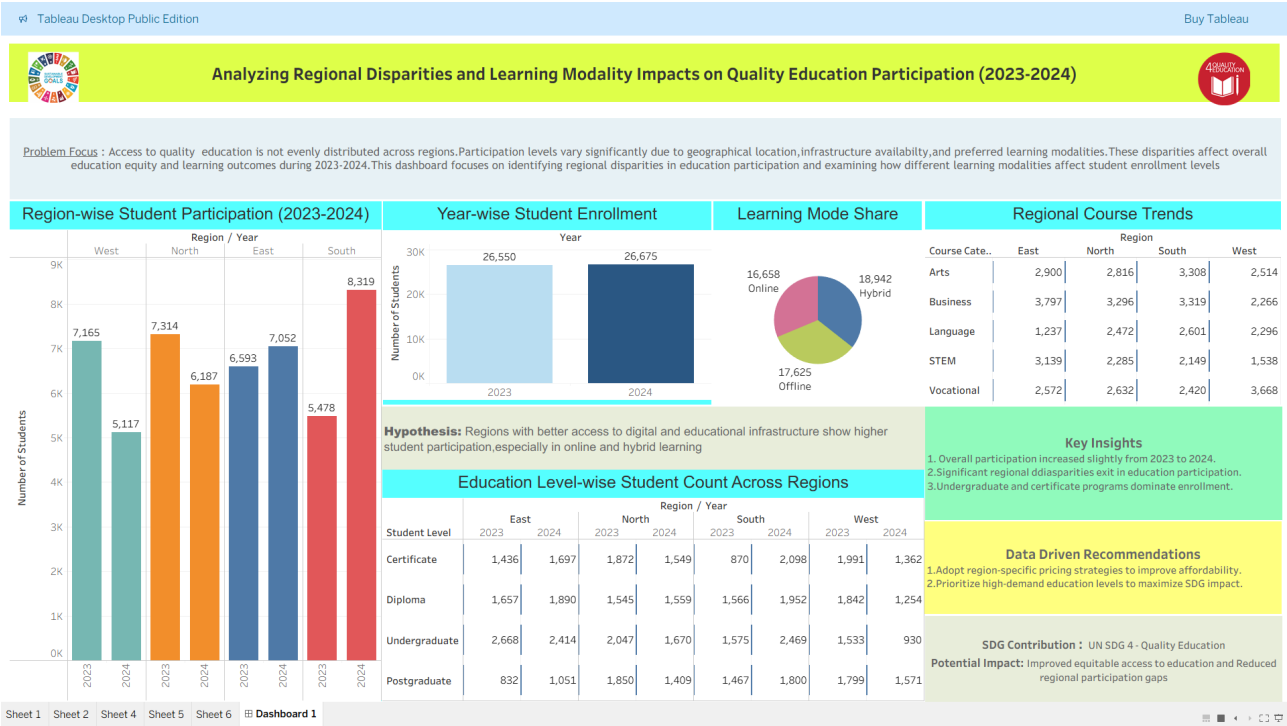


Analyzing Regional Disparities and Learning Modality Impacts on Quality Education Participation (2023–2024)

A data-driven dashboard study aligned with UN SDG 4 – Quality Education



Project Objective

To examine regional differences in student participation, compare annual enrollment, understand learning-mode distribution, and identify course and education-level patterns that can support more equitable education planning.

Project type: Tableau data visualization & analytical dashboard

Study period: 2023–2024

Theme: Quality Education, regional disparity, learning modality

Primary output: Interactive Tableau dashboard + analytical report

1. Executive Summary

The dashboard presents a two-year view of student participation across four regions—East, North, South and West—and combines regional, temporal, learning-mode, course-category and education-level perspectives. The dataset contains **53,225 student participation records across the two years**, with 26,550 in 2023 and 26,675 in 2024.

Overall participation increased by **125 students (0.47%)** from 2023 to 2024. However, the overall increase masks substantial regional changes: South grew strongly, East also increased, while North and West declined.

Metric	2023	2024	Change
Total participation	26,550	26,675	+125 (+0.47%)
South	5,478	8,319	+2,841 (+51.86%)
East	6,593	7,052	+459 (+6.96%)
North	7,314	6,187	-1,127 (-15.41%)
West	7,165	5,117	-2,048 (-28.58%)

Key interpretation

The dashboard indicates that overall participation was almost stable year-over-year, but the composition of participation changed considerably by region. This makes regional analysis more informative than looking only at the overall total.

- **Learning mode:** Hybrid accounts for 35.59% of the displayed participation, followed by Offline at 33.11% and Online at 31.30%.
- **Course categories:** Business has the highest combined participation, followed by Arts and Vocational.
- **Education levels:** Undergraduate participation is the largest aggregate level in the displayed education-level table.

2. Project Scope & Methodology

This project uses a Tableau dashboard to transform structured student-participation data into interactive visual evidence. The analysis is descriptive: it identifies patterns, differences and changes visible in the supplied dataset rather than claiming causal relationships.

Analytical dimensions

- **Time:** Compare student participation between 2023 and 2024.
- **Geography:** Compare East, North, South and West.
- **Learning modality:** Examine Online, Offline and Hybrid participation.
- **Course category:** Compare Arts, Business, Language, STEM and Vocational categories.
- **Education level:** Compare Certificate, Diploma, Undergraduate and Postgraduate participation.
- **SDG alignment:** Interpret the dashboard in the context of equitable access to quality education.

Important analytical note

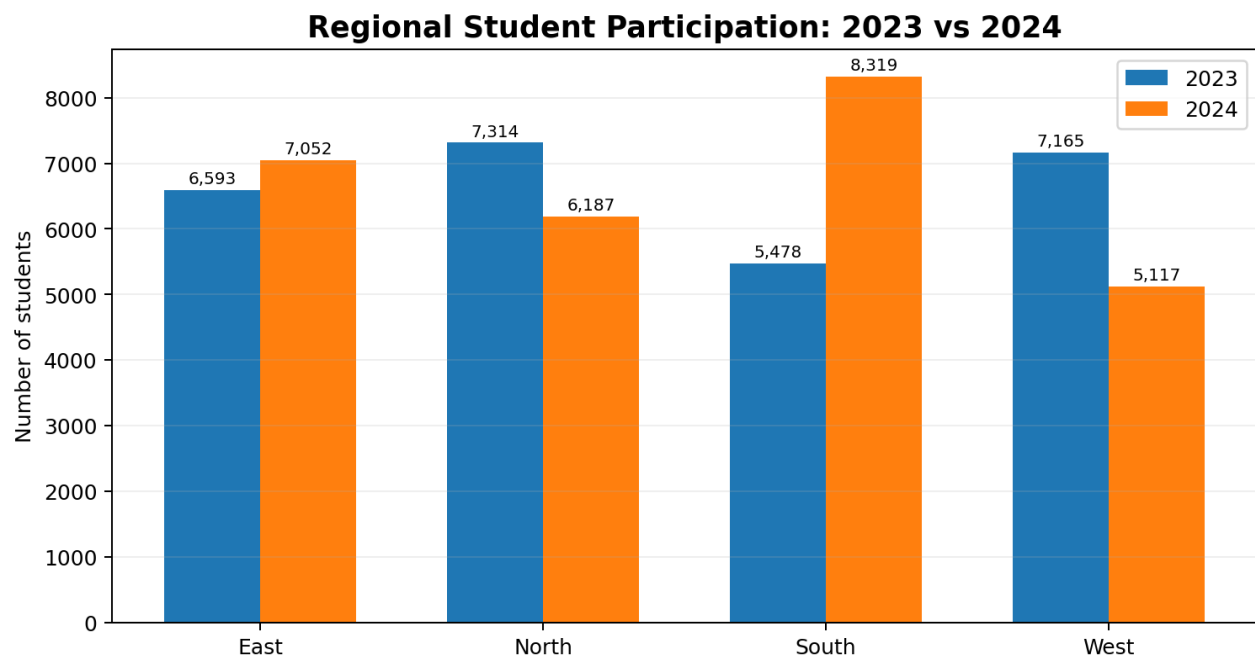
Hypothesis in the dashboard

The dashboard states the hypothesis that regions with better access to digital and educational infrastructure may show higher participation, especially in online and hybrid learning. The current dashboard does **not** display a direct infrastructure variable, so this hypothesis cannot be statistically tested from the shown data alone. The dashboard can reveal participation patterns that motivate further investigation, but it should not be interpreted as proof of causation.

Dashboard design approach

Component	Purpose
Regional bar chart	Shows 2023–2024 participation by region.
Year-wise enrollment	Shows overall annual participation.
Learning-mode pie chart	Shows Online/Offline/Hybrid composition.
Regional course table	Enables category-by-region comparison.
Education-level table	Shows level-wise participation by region and year.
Insights & recommendations	Converts descriptive findings into planning considerations.

3. Regional Participation Analysis



The regional comparison is the strongest evidence of disparity in the dashboard. In 2023, North recorded 7,314 students and West 7,165, while South had 5,478. In 2024, the pattern changed sharply: South reached 8,319 and became the largest regional total shown, whereas West fell to 5,117.

South — +2,841 students (+51.86%)

The largest increase in the dataset. The dashboard shows a major rise across the displayed education levels.

East — +459 students (+6.96%)

A moderate increase, with total participation moving from 6,593 to 7,052.

North — -1,127 students (-15.41%)

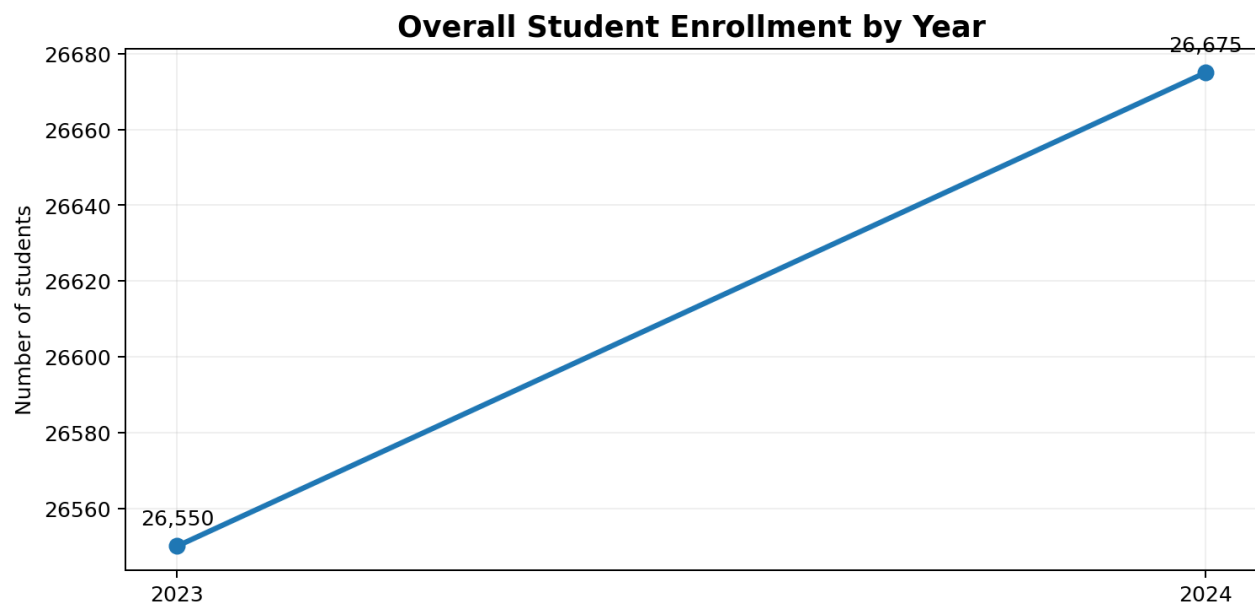
Participation declined from 7,314 to 6,187.

West — -2,048 students (-28.58%)

The largest percentage decline among the four regions.

Interpretation: The regional results suggest that a single national or overall enrollment number would hide meaningful geographic shifts. For planning purposes, regions with declining participation may require a different intervention mix from regions experiencing rapid growth.

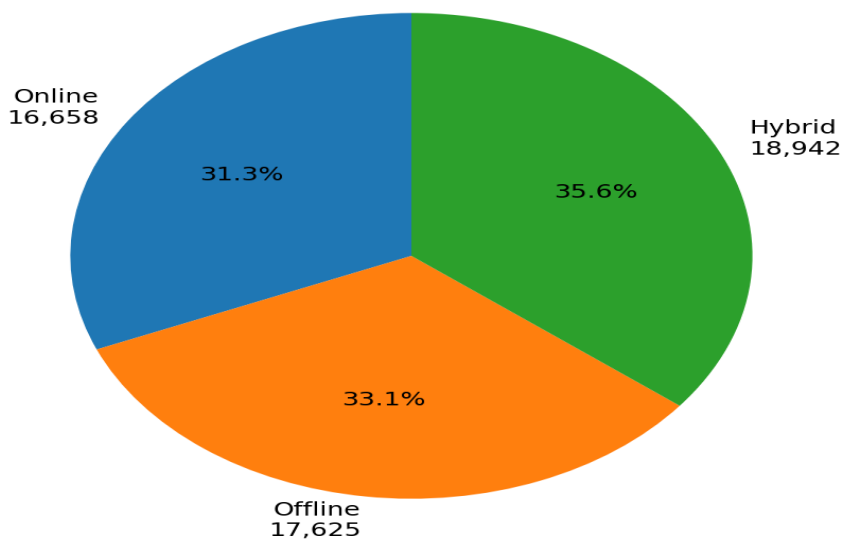
4. Year-wise Enrollment & Learning Mode



Year-over-year finding

Total participation moved from 26,550 in 2023 to 26,675 in 2024—an increase of only 125 students, or approximately 0.47%. This indicates broad stability at the aggregate level despite substantial regional redistribution.

Learning Mode Share

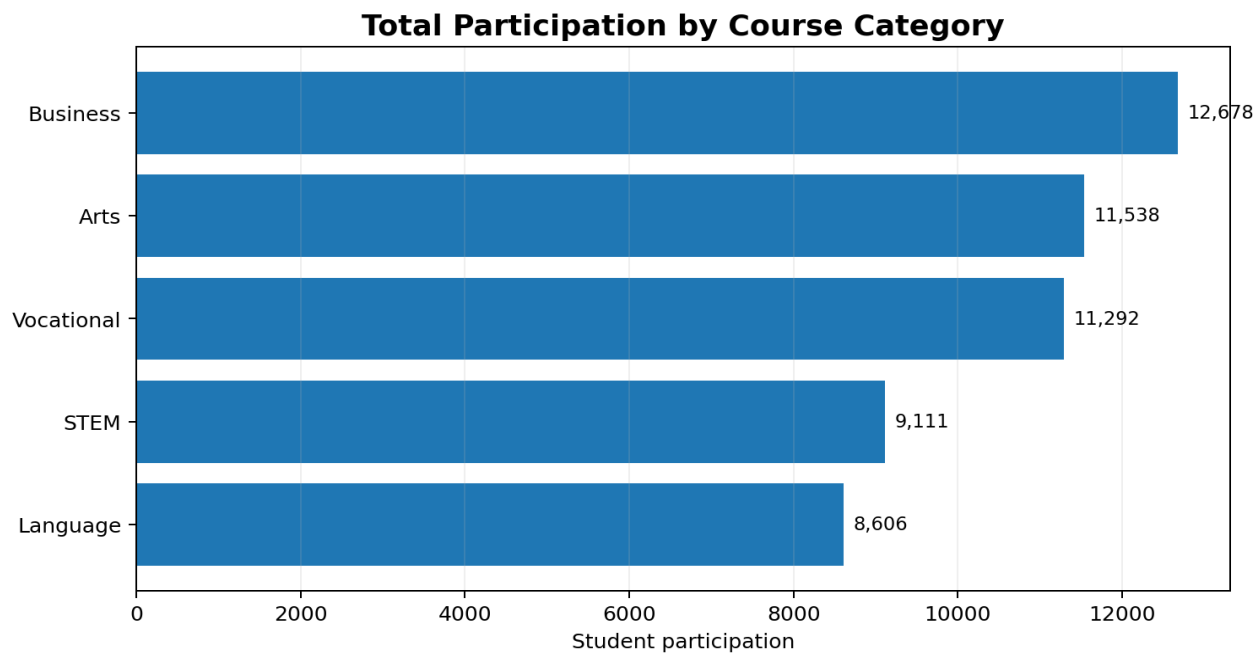


Learning-mode interpretation

- **Hybrid:** 18,942 students (35.59%) — the largest share.
- **Offline:** 17,625 students (33.11%) — a substantial traditional-learning component.
- **Online:** 16,658 students (31.30%) — nearly one-third of participation.

The relatively close shares across the three learning modes indicate that education participation is distributed across multiple delivery formats rather than being concentrated in a single mode.

5. Regional Course Trends



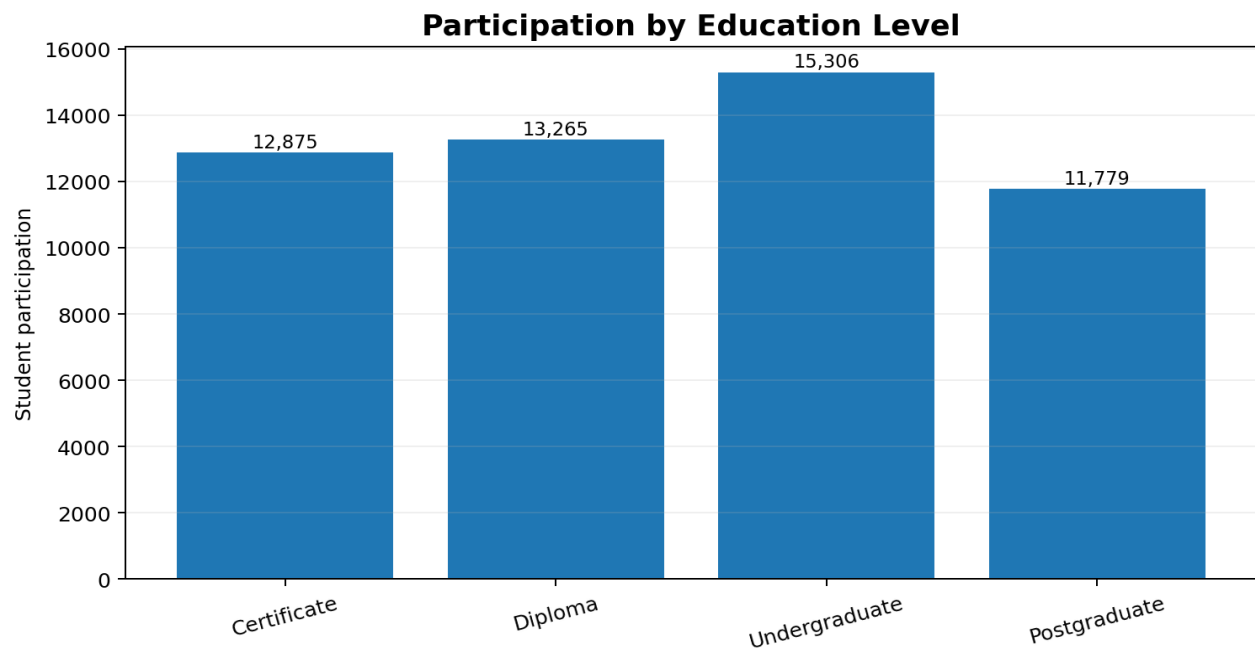
Across the four regions and both years represented by the dashboard, Business records the highest combined course participation at **12,678**. Arts follows at **11,538**, while Vocational records **11,292**. The category totals should be interpreted as participation counts in the supplied dashboard rather than as measures of course quality or student outcomes.

Course category	East	North	South	West	Total
Arts	2,900	2,816	3,308	2,514	11,538
Business	3,797	3,296	3,319	2,266	12,678
Language	1,237	2,472	2,601	2,296	8,606
STEM	3,139	2,285	2,149	1,538	9,111
Vocational	2,572	2,632	2,420	3,668	11,292

Notable regional patterns

- South has the highest Arts and Business participation among the displayed regional course counts.
- West has the highest Vocational participation, at 3,668, which is higher than its other displayed course categories.
- East has the highest STEM participation among the four regions, at 3,139.
- North has comparatively strong Language participation at 2,472.

6. Education Level-wise Analysis



Across the education-level table, Undergraduate participation is the largest aggregate category at **15,306**, followed by Diploma (13,265), Certificate (12,875) and Postgraduate (11,779).

Regional/year observations

- South shows a strong increase across all four education levels from 2023 to 2024.
- West records declines in Certificate, Diploma and Undergraduate participation, while Postgraduate participation is comparatively more stable.
- East increases in Certificate, Diploma and Postgraduate participation but shows a decline in Undergraduate participation.
- North shows declines across all four education levels in 2024 compared with 2023.

Planning implication

Education-level changes should be examined alongside regional changes. A region may require different capacity planning depending on whether growth is concentrated in certificate, diploma, undergraduate or postgraduate participation.

7. Key Findings

01	Overall participation is stable The total rises only 0.47% from 2023 to 2024, so the headline total changes very little.
02	Regional disparity is substantial South increases by 51.86%, while West decreases by 28.58%; the direction and scale of change differ strongly by region.
03	Learning modes are diversified Online, Offline and Hybrid each represent a meaningful portion of the total, with Hybrid having the largest displayed share.
04	Business has the highest course participation The Business category has the largest combined participation across the regional course table.
05	Undergraduate is the largest education level Undergraduate participation is the largest aggregate level in the education-level table.
06	Growth is not evenly distributed The small overall increase conceals major regional shifts, highlighting the value of disaggregated dashboard analysis.

8. Data-driven Recommendations

The following recommendations are derived from the observed dashboard patterns. They are planning considerations rather than claims that a specific intervention will automatically produce a particular outcome.

Regional targeting

Develop region-specific education participation strategies instead of applying one uniform approach across all regions.

Support declining regions

Investigate the reasons behind the North and West declines before allocating future capacity or outreach resources.

Learn from rapid growth

Study the drivers of South's strong increase and determine whether its growth is associated with capacity, access, awareness, program availability or other factors.

Maintain modality choice

Continue supporting Online, Offline and Hybrid pathways because all three account for substantial participation in the displayed dataset.

Prioritize demand-aware programs

Use course and education-level participation patterns to inform capacity planning, while avoiding the assumption that high enrollment automatically means high quality.

Add infrastructure variables

To test the dashboard hypothesis more rigorously, future versions should include internet access, device availability, institution capacity, distance to institutions and other measurable access indicators.

9. Limitations, Future Scope & Conclusion

Limitations

- The displayed dashboard is primarily descriptive and does not establish causal relationships.
- Infrastructure/access variables are referenced in the hypothesis but are not visibly included as measured fields in the dashboard.
- Only two years are shown, so longer-term trends cannot be established from this dashboard alone.
- Participation counts do not by themselves measure learning quality, completion, achievement or student satisfaction.
- The report assumes that the dashboard values are the authoritative project dataset supplied by the user.

Future scope

- Add 3–5 years of historical data for stronger trend analysis.
- Add socioeconomic, infrastructure and accessibility indicators.
- Add filters for gender, institution type, urban/rural location and other available dimensions.
- Introduce calculated fields such as year-over-year growth rate, regional share and course-level growth.
- Use statistical tests or regression models if sufficient explanatory variables are collected.
- Publish the Tableau workbook and cleaned dataset alongside the report in a GitHub repository.

Conclusion

The Tableau dashboard demonstrates how a compact education dataset can reveal patterns that are not visible from overall totals alone. The most important analytical message is the contrast between near-stable total participation and strong regional redistribution. Learning modality, course category and education-level views provide additional context for understanding participation. With additional years and measurable access/infrastructure variables, the project can be extended from descriptive visualization toward deeper statistical analysis.

10. GitHub README / Repository Structure Suggestion

File / Folder	Purpose
README.md	Short project overview, objectives, dashboard preview and key findings.
dashboard/	Tableau workbook (.twb/.twbx) and related packaged assets where appropriate.
data/	Source Excel/CSV dataset used for the dashboard.
report/	This detailed PDF project report.
images/	Dashboard screenshot and selected chart images.

Suggested repository title: Analyzing-Regional-Disparities-Quality-Education-Tableau

Suggested report filename: Tableau_Quality_Education_Dashboard_Project_Report.pdf